From reads to pair evidence

CLASP-seq | Analysis guide and reproducible example specification

Working document. Red **TO COMPLETE** fields identify missing or conflicting author inputs. Complete the fields affecting a procedure before using it at the bench. Recorded settings are not a claim of experimental optimization.

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01 What this guide will let a receiving lab reproduce

The analysis connects sequencing reads to sample, partition and construct identities, then assesses evidence for TCR-HLA-peptide associations. A useful release lets a receiving lab trace a reported association back through those steps and inspect the observations that were excluded.

Author completion draft
The source does not include a runnable pipeline, validated read architecture, interaction-calling method or reference result. The stages and table fields below are a proposed reporting contract. Red fields identify the exact software, files and decisions needed to make the guide executable.

TO COMPLETE - C12
Insert the analysis repository and release identifier, installation instructions, environment lockfile or container digest, supported system requirements and a named example-data release. Supply measured runtime and hardware requirements for that example.

02 Assemble a traceable input set

Input	Required content
Run and sample manifest	Experiment, replicate and batch IDs; biological role; sample/channel identity; verified PCR branch-to-index mapping.
Raw signal and reads	POD5 and available basecalled reads; acquisition and basecalling metadata; file checksums and archive locations.
Construct reference	Versioned full sequences; TCR α/β identities; HLA allele, peptide and engineered display construct; primer coordinates and qualification records.
Capture and amplification records	10x chemistry and custom changes; sample allocation; primer versions; complete branch history.
Controls and configuration	Named reference materials, expected behavior and its evidence; analysis settings and the run used to select them.

Verify files against the acquisition archive and manifest before processing. Resolve missing or duplicate sample IDs, sequence references and index assignments explicitly. Keep a record of each correction.

TO COMPLETE - C08 / C12
Provide the machine-readable sample and construct manifests, validated index aliases and an input-validation command. Include the expected validation report and the conditions that stop processing.

03 Identify the sample and partition in each read

Basecall and separate the amplicon branches

Pin the basecaller, model, chemistry and processing settings. Demultiplex the input TCR, experimental TCR, experimental HLA-peptide and input HLA-peptide branches using the released index rules. Retain assigned, ambiguous and unassigned reads with stagewise counts.

TO COMPLETE - C11 / C12
Insert exact basecalling and demultiplexing commands, configuration files and versioned output filenames. Define index orientation and coordinates, matching/error rules, adapter handling and ambiguous or missing-index behavior.

Parse partition barcodes and any retained molecular identifiers

The custom PCR sample index identifies an amplicon branch. It is distinct from a 10x partition barcode and from a UMI. The presence, orientation and location of recoverable barcodes or UMIs must be established for the released construct and chemistry.

TO COMPLETE - C07 / C12
Supply an annotated read-architecture diagram and real example reads. Specify barcode coordinates, whitelist, correction and collision rules; establish whether UMIs are retained and how they are processed. Provide the parser command, configuration and expected parsing report.

Keep the reaction in the identity
Interpret a partition barcode within its sample and capture reaction. The two experimental PCR branches come from the same reaction. Identical barcode strings in the two independent input reactions do not establish a pair.

04 Assign constructs without forcing an identity

1. Match reads against the versioned construct reference and preserve the evidence used for each assignment.
2. Retain partial, unexpected, chimeric and indistinguishable candidate sequences according to the released rules. Do not discard their existence from the QC summary.
3. Recover the linked TCR α/β identity when supported. Record unresolved chains rather than supplying a missing partner.
4. Report the actual HLA allele, peptide and engineered display construct. Preserve ambiguity between constructs that the observed sequence cannot distinguish.

TO COMPLETE - C12
Insert the construct-assignment command and reference files. Define error handling, required sequence support, alignment or matching parameters, chain-linkage evidence and ambiguous-assignment policy. Attach expected assignment tables and stagewise counts for the example.

If validated UMIs are unavailable, describe support as reads. Amplified reads do not become independent molecules or biological observations simply because they have separate read identifiers.

05 Connect experimental evidence and measure input representation

Associate TCR and HLA-peptide evidence only within the same experimental sample, capture reaction and partition. Handle multiple candidate constructs and possible multiple-cell partitions using the predeclared rules. Preserve the alternatives and reasons for exclusions.

Analyze the input TCR and HLA-peptide populations separately to describe library representation. They help establish which identities were available and detectable. They do not directly measure background co-occurrence within the experimental preparation.

TO COMPLETE - C12
Supply the partition-association and input-representation commands, output schemas, detection limits and filtering rules. Define how multiple constructs, multiple cells and unresolved identities enter evidence tables or exclusions.

Keep observation separate from interpretation
Co-positive sorting, shared partitions and high read counts alone do not establish specific recognition. Use explicit states such as unobserved, ambiguous, failed QC and insufficient evidence where appropriate. An unobserved pair is not a verified nonbinder.

06 Assess associations against controls

The interaction assessment must state what statistic is reported, its direction and how it is interpreted. It must connect the background model, normalization and decision rule to the experimental controls and replicate design.

TO COMPLETE - C12 / C13
Supply the null/background model, normalization, statistic definition, threshold-selection procedure, replicate requirements and any multiple-testing correction. Insert exact commands, control identities, expected control results and acceptance criteria. Include the data used to qualify these decisions.

Select settings using a development or control run, then freeze them for the stated evaluation. Record later changes and rerun affected comparisons. Export all assessed candidates, ambiguity states and exclusions alongside accepted calls.

Keep model predictions in separate columns or artifacts from assay observations and assay-defined decisions. A DecoderTCR or DecoderTCR-Fold score is not a substitute for the experimental calling method or an experimental control.

07 Publish tables that preserve the evidence

The following fields are proposed for a portable release. They are not an existing API. Final schemas should specify field types, identifiers, units, missing-value rules and permitted status values.

Table	Proposed content
Sample manifest	experiment_id, replicate_id, batch_id, sample_id, sample_role, channel_id, sample_index_id, sample_index_sequence, library_manifest_version.
Library manifest	construct_id, library_id, construct_sequence_id, linked tcr_id or hla_peptide_id, full sequence references, intended identity and validation status.
Partition assignments	sample_id, channel_id, partition_barcode, candidate identities, read support, molecule support only if defined, assignment_status and exclusion_reason.
Pair evidence	Experiment and replicate, tcr_id, hla_peptide_id, supporting partitions, input representation, assay-defined statistic and decision, analysis version.
QC and exclusions	Counts entering and leaving each step; failed or ambiguous assignments; missing references; control results and reasons for exclusions.

Label every support count and denominator. Reads, molecules, partitions, sorted events and biological pairs are different quantities. Preserve the underlying assignment records so aggregate counts can be checked.

TO COMPLETE - C12
Supply the final schema files, schema-validation command and a correctly populated example for each table. Identify which columns are measured, calculated or interpretive.

08 Make one small example fully reproducible

Release a permitted example that covers the experimental and input branches, their references, and the controls needed to interpret the result. It should run without internal storage access. Preserve real failures and ambiguities rather than replacing them with idealized output.

Deliverable	Author insertion required
Example input bundle	TO COMPLETE: persistent data link, sample/reference files, raw or read data, checksums and reuse terms.
Execution record	TO COMPLETE: exact ordered commands, configuration, software/reference versions and pinned environment.
Intermediate outputs	TO COMPLETE: branch counts, parsing results, construct assignments, partition evidence and QC/exclusion tables.
Reference result	TO COMPLETE: pair-evidence table, control assessment, figure-generation command and expected figures.
Comparison method	TO COMPLETE: expected file checksums or justified numerical tolerances; measured runtime and hardware.

A reviewer should be able to follow one accepted association and one ambiguous or excluded candidate from the final table back to their supporting records. Recreate the figures from the released outputs; the source's uncaptioned result image is not yet a reproducibility target.

TO COMPLETE - C13 / C14
Record an outside operator's reproduction attempt, the bundle version, clarifications required and the resulting corrections. Add the release authors, contact, version, persistent identifier, licenses and change log. Describe independent reproduction only after it has occurred.

Source notes

Sources are the method authors' laboratory records, as transcribed in the supplied consolidation. The internal source files are not included in this download.

S1. CLASP-seq: the Cell_x_Cell TCR-pMHC pairing protocol. Author-supplied source-checked draft, 11 September 2026. Sections 2.3-2.4, 6.1-6.4, 7.4, 8, 10-12. Analysis stages and portable outputs are explicitly proposed in that draft; no analysis implementation or results are supplied.

Protocol and resources